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Shreyas Sundara Raman

Education

Master's 2024-2025 / Bachelor's 2019-2024 / GPA: 4.0

Brown University, *Master & Bachelor (Honors) of Computer Science, Magna Cum Laude*.

- Theses in Corrective Robot Planning (Sc.B.) and Factored Decision Making (Sc.M)
- Research courses: Robotics, Reinforcement Learning, SSL, LLM Interpretability, Deep Learning, Machine Learning, Databases
- Awards: Young Researcher - Heidelberg Laureate (2025), CS Senior Prize (2024), Sigma Xi (2024), KTR UTRA Award (2022)
- Teaching: Head TA for AI [120 students], Meiklejohn Mentor, STEMS Tutor

Research & Engineering Experience

2025-Present — **Software Engineer, Machine Learning** End-to-End Autonomy & Representation Learning,
Nuro, led by Shuvam Chakraborty.

- Cross-Attention Representation Learning: Designed and implemented **DETR positional encodings + fixed query cross-attention** in PyTorch/C++ that learned vehicle/curb-specific attention representations, which **increased frequency of nudges around unmapped vehicles/debris by 30%**; resolved **FP16 OOM/NaN failures** via **mixed-precision graph segmentation**.
- Synthetic Closed-Loop Learning: Led integration of **3D Gaussian Splatting** synthetic sensor data into closed-loop training/evaluation that **eliminated traffic-light reaction / stuck failures** and improved **occlusion creeping by 15%**. Aligned a **100× synthetic-to-real radar/lidar scale mismatch** causing training NaNs by correcting for ego-motion compensation.
- Cross-Platform Representation Adaptation: ideated and developed a **canonicalized radar representation** using **cross-platform adapters** in **PyTorch** that aligned platform-specific heterogeneous radar power/azimuth distributions, which **enabled training across 3 platforms for an on-road release** and improved **cross-platform generalization by 20%**.
- Auxiliary Representation Tasks: adapted and implemented a **DETR-style auxiliary head** for lane polyline/type reconstruction from BEV tokens **in PyTorch/TensorFlow** that identified bottlenecks from **19% solid-vs-dashed boundary misclassification** and reduced **lane-favoring failures by 21%**
- Large-Scale Data Curriculum: fixed a **40% train/eval geometry-loss gap** caused by rare-data oversampling, empirically finding that **1:7 data curriculum** following geometric sampling unblocked **2.5×-scale training** for on-road deployment.

2023-2025 — **Research Engineer** Robot & Representation Learning,
Brown University, Intelligent Robots Lab (IRL), led by Prof. George Konidaris.

CoRL 2024 — Symbolic Skill Abstraction: co-first author, LEAP @ CoRL 2024, Best Poster @ Breaking Results ICRA 2025 Dec/24 - Jun/25

- Developed a **VLM-based method in C++ and PyTorch** to **learn symbolic descriptions of robot skills** from visual observations, achieving **75% success rate** (6% less than expert) within 5 refinement iterations and improving **sim-to-real transfer** across AI2Thor and **Isaac Sim**
- Implemented and evaluated **4 contact-rich manipulation skills** ('pick up', 'place', 'pour', 'stir') using a general purpose **ROS service** to **replay end-effector trajectories** on **2 robots** (SPOT and Franka), adopted by **10+ researchers**

Self-Supervised Factorized Representations for Embodied Agents: 1st-author Jan/23 - Jun/25

- Designed a **self-supervised objective and architecture** with **LeanRL** and **PPO** to disentangle independently changing factors relevant to decision-making, yielding **30% more generalizable policies** than **DreamerV2** on MiniGrid and ProcGen
- Built distributed SLURM training infrastructure for 4M-step RL experiments, **reducing training time 3×** and instrumented **representation collapse and learning dynamics** with W&B; prepared the training stack into a **PyPI library**

2022-2024 — **Research Engineer** Vision-Language Reasoning & Robot Planning,
Brown University, Humans-to-Robots Lab, led by Prof. Stefanie Tellex.

ICRA 2024 — CAPE: Visual Language Reasoning for Robot Planning: 1st author, Oral Presentation, 250+ citations Jan/22 - Sep/24

- Proposed a **novel self-corrective planning system** in Python that recovers from PDDL precondition failures by combining **language feedback using LLMs** (GPT-3.5) with **CLIP**, **CLIPSeg** and **SLAM** to localize objects from raw images.
- Reduced **corrective interventions by 10×** and **improved correctness by 20%** with **>100× faster** planning time compared to the SoTA (SayCan) across 750 household tasks on VirtualHome simulator and the **SPOT Robot**.
- Developed a novel **RoBERTa embedding** based **action-grounding module** to map unstructured LLM outputs to executable robot actions, **improving plan correctness by additional 6%**

ICRA 2024 — Safety Chip: Constraints for LLM-driven Robot Agents: 2nd author, Oral Presentation, 150+ citations Jan/23 - Sep/24

- Developed method to **represent open-ended language constraints as LTL-automata** with a **finetuned LLM (GPT-3.5)** in Python, **improving restriction adherence by 15-40%** over prior work on the SPOT, for up to 5 simultaneous constraints

Additional Engineering Experience

- Developed a **mobile-manipulation** skill hosted on **Docker** for SPOT with **ROS Noetic, rospy, tf2 and C** and **RViz/MoveIt!** for trajectory visualization as part of an **open-source skill library** for SPOT

2023-2024 — **Reinforcement Learning Research Engineer**,
Brown University, Brown Integrative General AI Group (BigAI), led by Prof. Michael Littman.

RLC 2024 — Tiered Reward May/23 - Mar/24

- Built **distributed RL training** pipeline across 3 MiniGrid environments and **500+ runs** to evaluate reward-design hypotheses using **PyTorch** and **PPO/Stable-Baselines3**
- Developed and evaluated an environment-independent reward structure with **Pareto-optimal goal visitations guarantees** using PyTorch/PPO, enabling **3× faster policy convergence** than standard reward-shaping.

2020-2021 **Computer Vision Research Engineer**,
Brown University, Serre Labs, led by Prof. Thomas Serre.

CogSci 2025

- Visual Attention of Dogs: 2nd-author, Oral Presentation @ CV4Smalls, WACV 2024 May/20 - Dec/21
  - Engineered a **full-stack segmentation pipeline** in **TensorFlow** (data processing, **MaskRCNN fine-tuning**, and benchmarking) to predict dog gaze fixations from live video streams, achieving a **median 60% IoU** and **75% accuracy** on unseen videos

Engineering Projects

SLANG: SPOT & Language, [GitHub][Presentation][Report].

- Developed a **language-conditioned grasping system** enabling SPOT to identify and grasp objects from **semantic descriptions** (e.g., "something healthy to eat") using **CLIPSeg** and dynamic **Open3D point clouds**, improving **grasp success by 10%** and reducing **wall-time by 4**× versus ViLD

Autonomous Raspberry Pi Drone, [GitHub][Demo].

- Engineered **flight software** for a drone **from scratch** using **AutoCAD, ROS/ROSPy** hosted on a **Raspberry Pi**
- Implemented **5 modules** with **1000 LOC** including autonomous altitude control with **PID**, velocity estimation with **optical flow** and localization with **SLAM** using **Unscented Kalman-Filters**

LLM Interpretability for State Tracking, [GitHub][Report].

- Built systems to **benchmark LLMs state-tracking** ability across 3 logic tasks and 1000+ counterfactual prompt pairs.
- Implemented **residual stream and attention head patching** using TransformerLens to **identify consistent layer/head circuits** responsible for state-tracking, across **8 GPT/Pythia models** ranging **8M–1.5B** parameters.

ChromaLock: Keypad Lock Embedded System, [GitHub][Proposal][Report].

- Built a password-protected keypad lock in **C++** using **Arduino**, **PWM servo motors**, and an LCD interface, with watchdog timers for idle auto-locking, achieving **<2s lock latency** and **>95% test coverage** via an **automated CI/CD pipeline**.
- Ran **A/B testing** on system usability and safety, with test group showing **20% higher satisfaction**